How much edit experience should the model receive?

Sequence-edit JEPA validity audit and representation decision

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The task is synthetic typo repair

- Official iGSM supplies a problem and its clean, multi-step worked solution.
- We damage that solution with token insertions, deletions, and replacements.
- The expert path is the exact reverse of those corruptions, so it is oracle denoising rather than naturally observed editing.
- More independent damaged solutions add breadth; more alternative edits per state add local action coverage.

Why it matters: we must establish healthy primitive dynamics before asking whether hierarchy helps.

The old hierarchy round is invalid

- All five jobs failed before their first optimizer step, so they provide no architectural evidence.
- The immediate symptom was a 276-token “sentence” exceeding a 128-position table.
- The cause was deeper: official steps often end in fused tokens such as `6.`, while the adapter split only on standalone `.`.
- The corrected adapter preserves official nested steps and exactly reverses every audited corruption.

What to notice: this is an invalid result, not evidence against hierarchy.

The corrected data contract passes its first gate

Audit check	Observed result
Exact terminal recovery	128/128
Multi-step solution collapse	0/128
Official solution length	median 166.5 tokens
Official steps	median 9
Maximum step length	p95 43 tokens
Repairs per trajectory	mean 11.14
Repair/minimum-distance ratio	mean 1.029

Why it matters: the model now sees the intended step sequence, but these are data-validity observations, not trained-model results.

All six process-valid cells ignore the edit action

Cell	Prediction	No change	Shuffled/matched
512 trajectories	.397	.575	1.00014
2,000 trajectories	.394	.302	1.00000
6,000 trajectories	.224	.113	1.00005
K=1 at 2,000	.287	.268	1.00003
K=4 at 2,000	.175	.123	1.00003

What to notice: lower raw error does not establish dynamics; most cells lose to copying the current state, and shuffling the action changes nothing.

One-token effects disappear in the global target

- Four exact global next-buffer targets from one state are separated by only .000228 normalized L1.
- The exact changed-step targets for those edits are separated by .634, about 2,775 times more.
- The old K=4 model assigns changed-step outcomes at 24.7%, chance for four alternatives.
- The repair predicts exact changed-step anchors for expert and mechanical alternative edits, with no quality or preference labels.
- Coefficients .25, 1, and 4 plus a lower-learning-rate check all completed cleanly.

Question: does stronger local supervision make the pooled predictor use the action?

Local supervision still leaves the model action-blind

Weight	Pred.	Persist.	Shuf./match	Assignment
.25	.173	.130	1.00004	24.92%
1	.173	.146	1.00003	24.85%
4	.186	.150	1.00151	25.04%
4, low LR	.354	.177	1.00041	25.00%

- Assignment stays at 25% chance; every cell loses to persistence.
- Rank falls 11.6–26.4%; no coefficient deserves confirmation.
- The “local” head still projects one pooled global-state prediction.

Next decision: one direct observed-step attention test under the unchanged gates; otherwise retire this state family.

Direct step attention also fails the causal gates

Gate	Observed result
Local K=4 assignment $> 35\%$	27.56%
Shuffled/matched error ≥ 1.05	0.99933
Prediction beats persistence	.123 vs .090
Effective-rank loss $\leq 10\%$	23.4%

- Direct observed-step access creates some local separation, but not reliable action use.
- The recursive score is unassessed: this branch has no rollout API, and the dormant core is not a substitute.
- No coefficient, K/data, hierarchy, rollout, or LDAD continuation is justified.

Decision: retire this state family; require a token-aligned recursive CPU fixture before any restart.